ESD (After Endsem):

RCC and Clock Control:

Crystal Oscillator is used to generate clock signal.

The **Reset and Clock Control (RCC)** unit is crucial for managing:

- System and peripheral clocks
- Multiple reset types (system, power, RTC domain)
- Power-efficient clocking strategies

RCC in STM32G0 builds on STM32F0 with **new features** such as:

- More flexible NRST pin modes
- Better PLL structure with 3 post-dividers (PLLPCLK, PLLQCLK, PLLRCLK)

*Types of Resets

1. **System Reset**
    - Resets most registers but retains RTC
    - Triggered by: NRST pin, watchdogs, software request, low-power security fault, option byte loading, brown-out, power-on
2. **Power Reset**
    - Triggered by **Brown-Out Reset (BOR)**
    - RTC domain (powered by VBAT) retains contents unless explicitly reset
3. **RTC Domain Reset**
    - Controlled via `BDRST` bit

    - Occurs when both VDD and VBAT have been powered off

***Internal Oscillators:**

- **HSI16**: 16 MHz, trimmed in factory, 1% accuracy, fast startup
- **LSI**: 32 kHz RC oscillator, ultra-low power (~110 nA), RTC and watchdogs

***External Oscillators:**

- **HSE**: 4–48 MHz, external crystal/resonator, clock security system
- **LSE**: 32.768 kHz crystal/resonator, used for RTC, low-power timers

***Phase-Locked Loop (PLL)**

- Input from HSI16 or HSE
- Outputs: PLLPCLK, PLLQCLK, PLLRCLK
- Useful for different peripherals at different frequencies
- Multiplies clock like 8Mhz -> 16Mhz
- Controlled using Voltage controlled oscillator (VCO)

```
### Naming Convention

- HS: High speed
  - range: MHz
- LS: Low speed
  - range: KHz
- ..E: Low speed external
- ..I: Internal speed external

### Precision

- where timing precision is not very critical (~1 second delay), there we use internal clocks
```

  - 10-20% tolerance
- External clocks are very very precise (~3 ppm = 3 points delay in a million units)

- LSE and LSI go to RTC
- HSI and HSE go for microcontroller operation

- Internal Clock: 8-16MHz
- HSI16 = 16Mhz high speed clock
  - 1% error
  - I2C, USART, LPUART, CEC and UCPD can enable the HSI16
  - over [0-80C] : +- 1%
  - startup time: 1us
- HSE = 4 - 48MHz
  - connected with OSC_IN and OSC_OUT
  - To make the clock synthesiser start to oscillate, we need to give feedback potential

Failsafe

- Clock security system
- CSS is used as a backup mechanism in case of clock failure in a MCU
- Note: MCU don't boot at all without clocks

- AHB: Advanced High Performance Bus
- APB: Advanced Peripheral Bus

RESET

- Resets all registers except certain RCC register and the RTC domain
- Low level on the NRST pin (external reset) : like the reset push button
- Windows Watchdog = WWDG
- IWDG = another watchdog
- Software reset (like the BSOD on windows)
- Brown-out or power-on reset if the voltage goes out of bounds (either too low or high)
- RCC_CSR register (kinda like a log why the system did a reset)

*Clock Frequency

- Max **SYSCLK = 64 MHz** (in Range 1 power mode)
- Derived from: HSI16, HSE, LSI, LSE, or PLL output
- Can be scaled down with **prescalers** for AHB and APB clocks
- Important buses:
  - **HCLK (AHB)**: Main system bus
  - **PCLK (APB)**: Peripheral bus

>>Power consumption and performance are directly tied to how clocks are configured.

*Calendar (RTC clock sources)

- RTC requires a **low-frequency, stable** clock.
- You can use:
  - **LSE (32.768 kHz)**: most accurate and ideal for timekeeping/calendar
  - **LSI (32 kHz)**: less accurate but low power
  - **HSE / 32**: possible but high power for a calendar function

Use LSE for RTC when calendar accuracy is important.

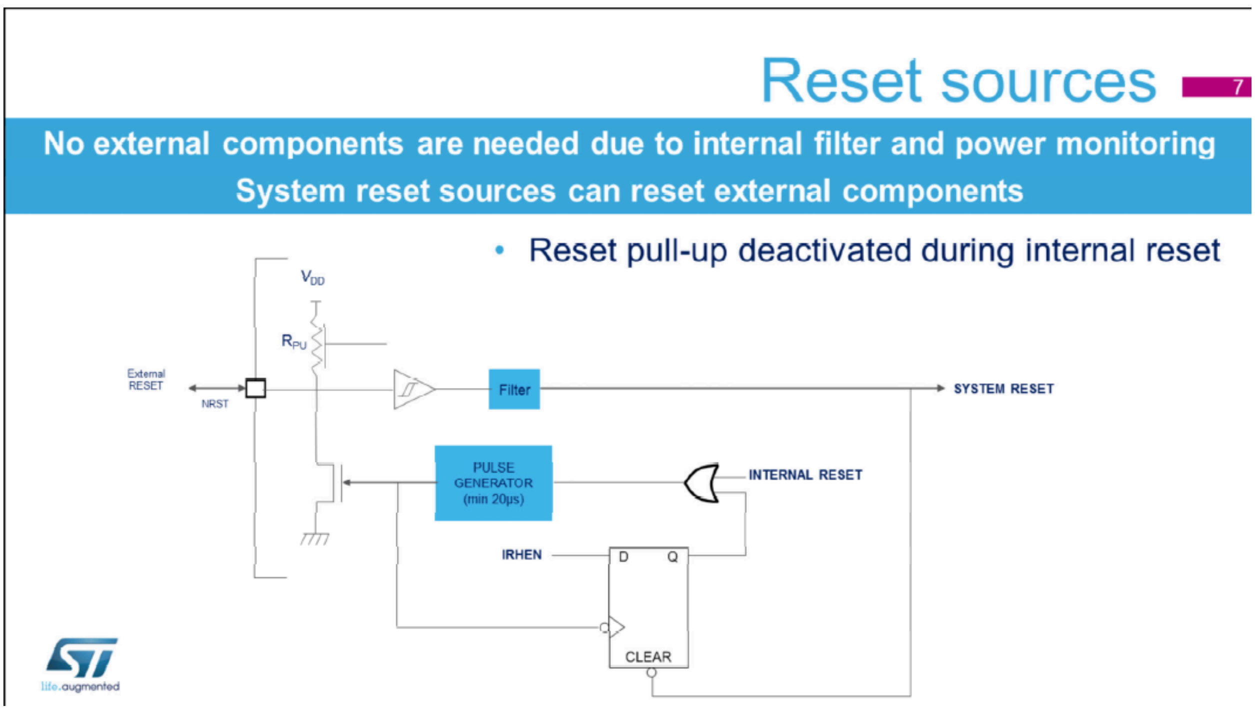
*Why 32.768 kHz is used in Digital Clocks

- 32,768 = 2^{15} → Easy to implement 1 Hz signal using binary counters.
- It allows RTCs to count seconds precisely using a **15-stage binary divider**.

***Reset Pull-Up Deactivated During Internal Reset**

- Normally, the **NRST pin** has an internal **pull-up resistor** to keep it HIGH when unused.
- **During an internal reset**, this pull-up is **disabled** to:
    - Reduce power draw
    - Allow better control if external circuitry is connected.

Also, **I/O pins are set to analog mode** during reset to avoid leakage current from Schmitt triggers. Clever for power savings!



Peripheral Clock Gating: Clocks to peripherals are disabled by default; enable only what's needed → saves power

Low-power modes: Most clocks disabled, but **LSI and LSE can stay ON** to keep time or monitor watchdogs